

Firm Level Variation Analysis

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Libraries

```
library(here)
library(readr)
library(dplyr)
library(ggplot2)
library(lubridate)
library(tidyr)
```

Load data

```
# Load FDA approval data

fda_approvals <- read_csv(
  here(
    "processed_data/fda_approvals/fda_sdud_pdl_merged.csv"
  )
) |>
rename(
  approvals = n_approvals,
  firm = sponsor_name
) |>
mutate(post = approval_year >= 2014) |>
filter(approval_year >= 2010)

# Load patent data

patents <- read_csv(
  here(
    "processed_data/patents/patents_sdud_pdl_merged.csv"
  )
) |>
rename(
  atc_level1 = ATC1,
  patents = num_patents) |>
mutate(post = patent_year >= 2014) |>
filter(patent_year >= 2010)
```

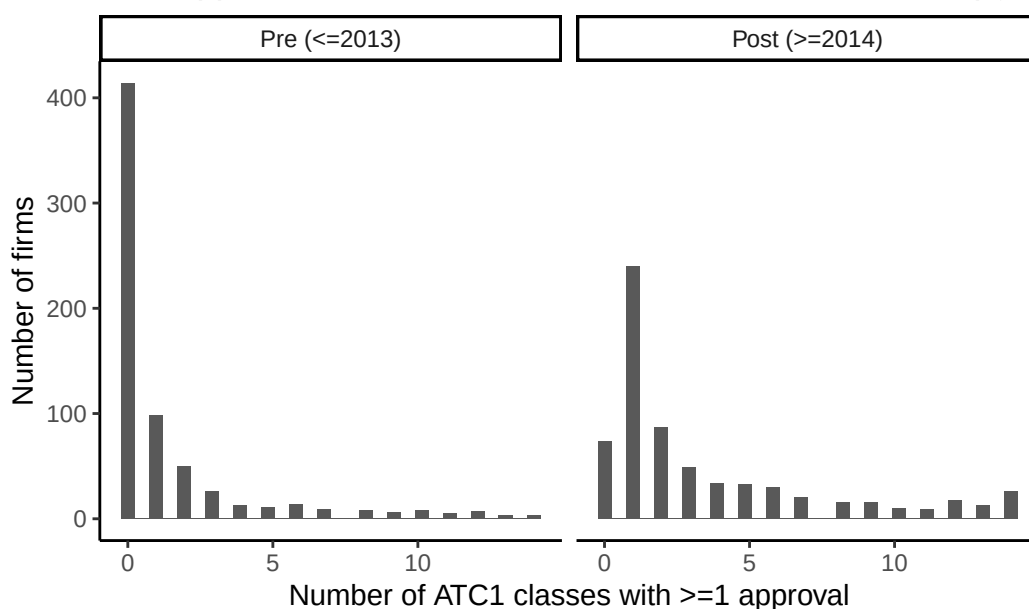
Between firm variation for ATCs

FDA

```
fda_firm_atc_prepost <- fda_approvals |>
  group_by(firm, atc_level1, post) |>
  summarise(any_innov = max(approvals > 0), .groups = "drop") |>
  group_by(firm, post) |>
  summarise(
    n_atc_innov = sum(any_innov),
    .groups = "drop"
  )

# Plot: FDA
ggplot(fda_firm_atc_prepost, aes(x = n_atc_innov)) +
  geom_histogram(bins = 30) +
  facet_wrap(~ post, labeller = as_labeller(c(`FALSE` = "Pre ( 2013)", `TRUE` = "Post ( 2014)"))
  labs(
    title = "FDA approvals: number of ATC1 classes with innovation (by firm)",
    x = "Number of ATC1 classes with 1 approval",
    y = "Number of firms"
  ) +
  theme_classic()
```

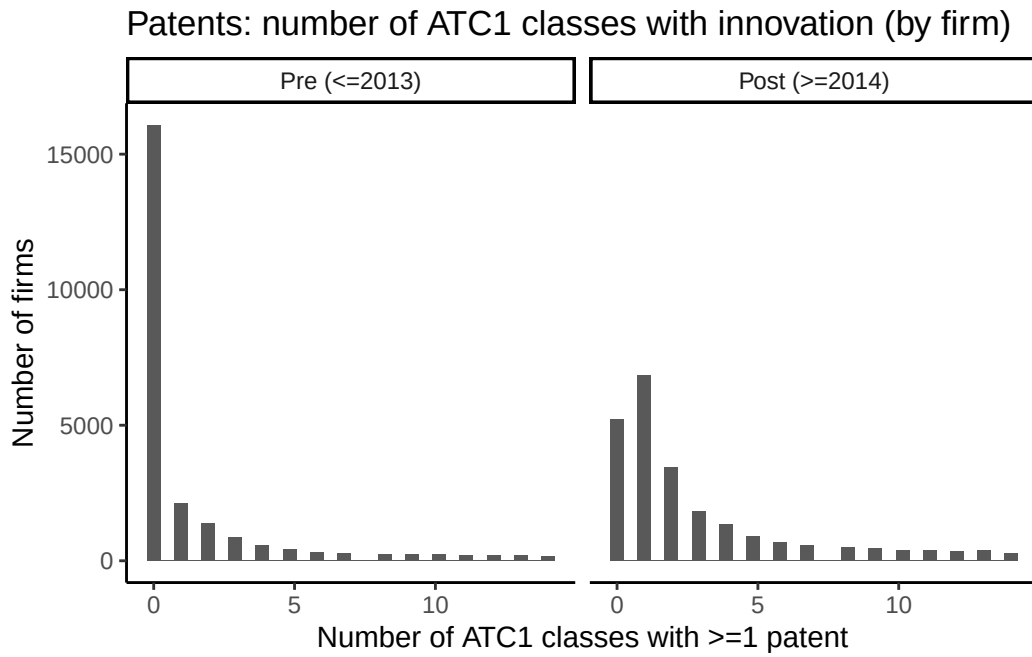

FDA approvals: number of ATC1 classes with innovation (by firm)



Patents

```
patent_firm_atc_prepost <- patents |>
  group_by(firm, atc_level1, post) |>
  summarise(any_innov = max(patents > 0), .groups = "drop") |>
  group_by(firm, post) |>
  summarise(
    n_atc_innov = sum(any_innov),
    .groups = "drop"
  )

ggplot(patent_firm_atc_prepost, aes(x = n_atc_innov)) +
  geom_histogram(bins = 30) +
  facet_wrap(~ post, labeller = as_labeller(c(`FALSE` = "Pre ( 2013)", `TRUE` = "Post ( 2014)"))
  labs(
    title = "Patents: number of ATC1 classes with innovation (by firm)",
    x = "Number of ATC1 classes with 1 patent",
    y = "Number of firms"
  ) +
  theme_classic()
```



Within firm variation

FDA

```
#| results: asis
#| fig-width: 10
#| fig-height: 6

fda_firm_atc_counts <- fda_approvals |>
  group_by(firm, atc_level1, post) |>
  summarise(approvals = sum(approvals, na.rm = TRUE), .groups = "drop") |>
  filter(approvals > 0)

fda_firm_index <- fda_firm_atc_counts |>
  group_by(firm) |>
  summarise(total_approvals = sum(approvals), .groups = "drop") |>
  arrange(desc(total_approvals)) |>
  mutate(firm_id = row_number())

fda_plot_data <- fda_firm_atc_counts |>
  left_join(fda_firm_index, by = "firm") |>
  pivot_wider(names_from = post, values_from = approvals, values_fill = 0) |>
  rename(pre = `FALSE`, post_val = `TRUE`)

atc_list_fda <- unique(fda_plot_data$atc_level1)

for (i in atc_list_fda) {

  single_atc_fda <- fda_plot_data |>
    filter(atc_level1 == i)

  p <- ggplot(single_atc_fda, aes(x = firm_id)) +
    geom_segment(aes(xend = firm_id, y = pre, yend = post_val),
      color = "grey80", alpha = 0.5) +
    geom_point(aes(y = pre, color = "Pre (<=2013)"), size = 1, alpha = 0.1) +
    geom_point(aes(y = post_val, color = "Post (>=2014)"), size = 1, alpha = 0.1) +
    scale_color_manual(values = c("Pre (<=2013)" = "#2c3e50", "Post (>=2014)" = "#e67e22")) +
    scale_y_continuous(labels = scales::comma) +
    labs(
```

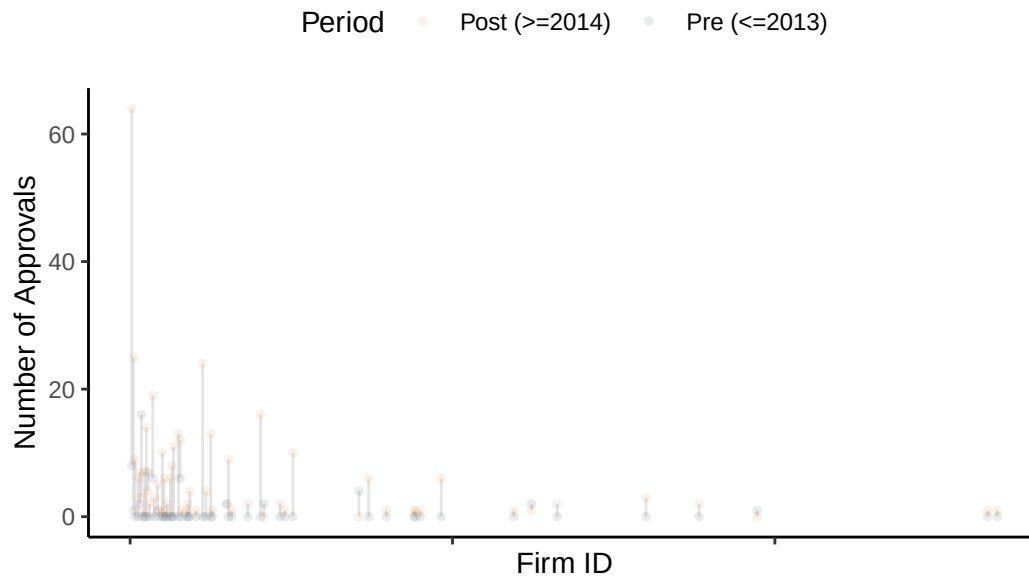
```

    title = paste("FDA Within-Firm Variation: ATC Level", i),
    x = "Firm ID",
    y = "Number of Approvals",
    color = "Period"
  ) +
  theme_classic() +
  theme(
    axis.text.x = element_blank(),
    legend.position = "top"
  )

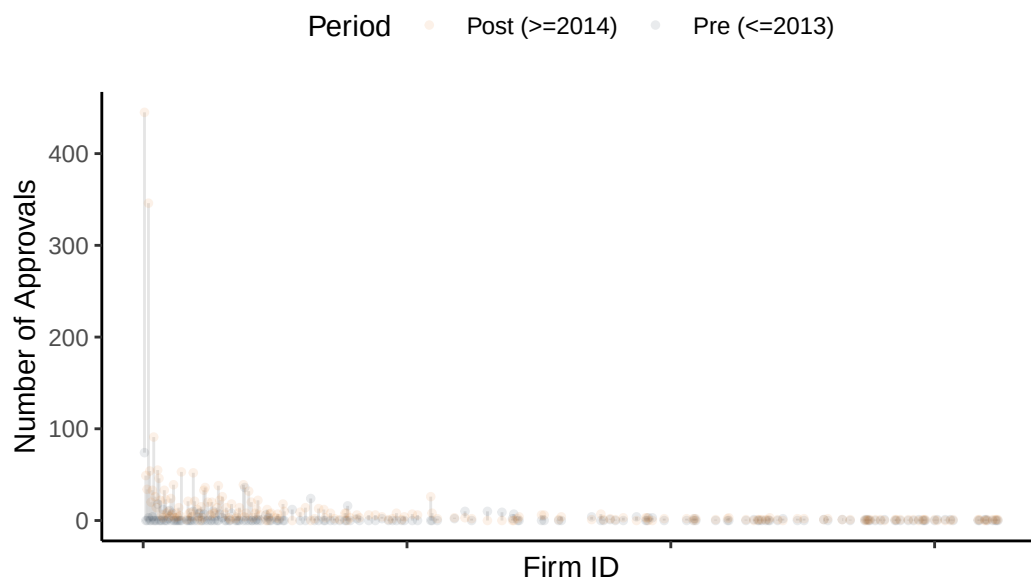
  print(p)
}

```

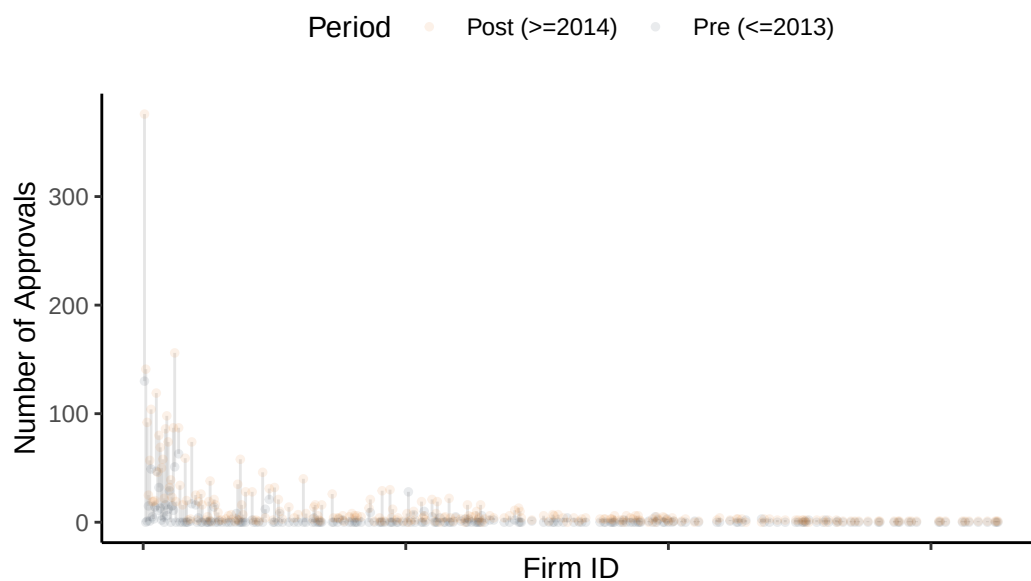
FDA Within-Firm Variation: ATC Level P



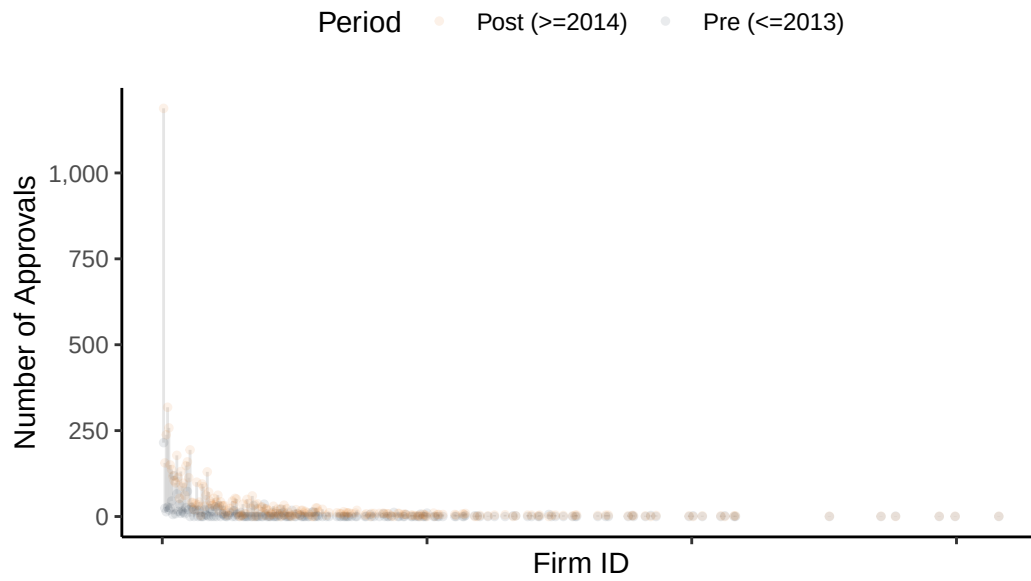
FDA Within-Firm Variation: ATC Level V



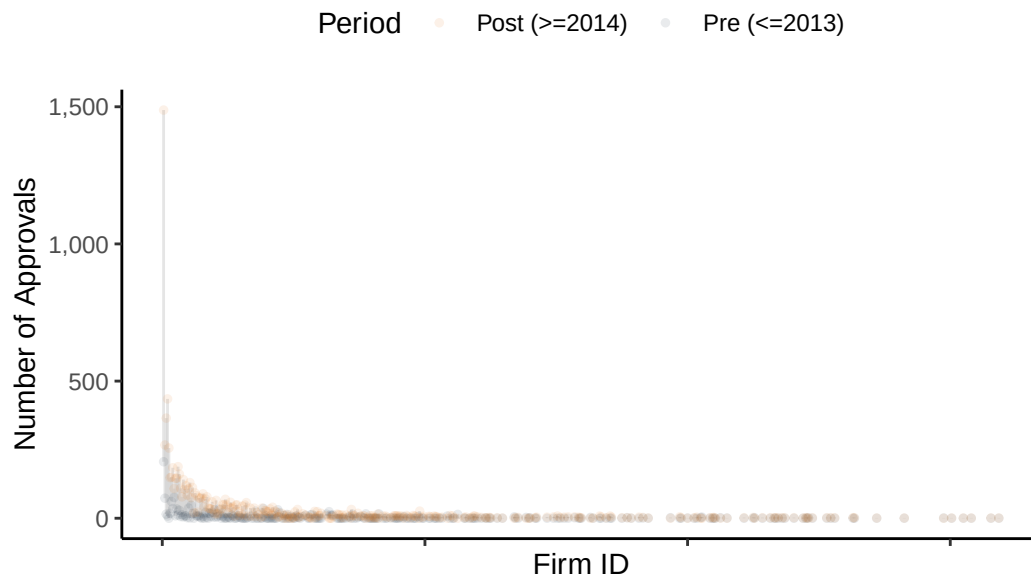
FDA Within-Firm Variation: ATC Level L



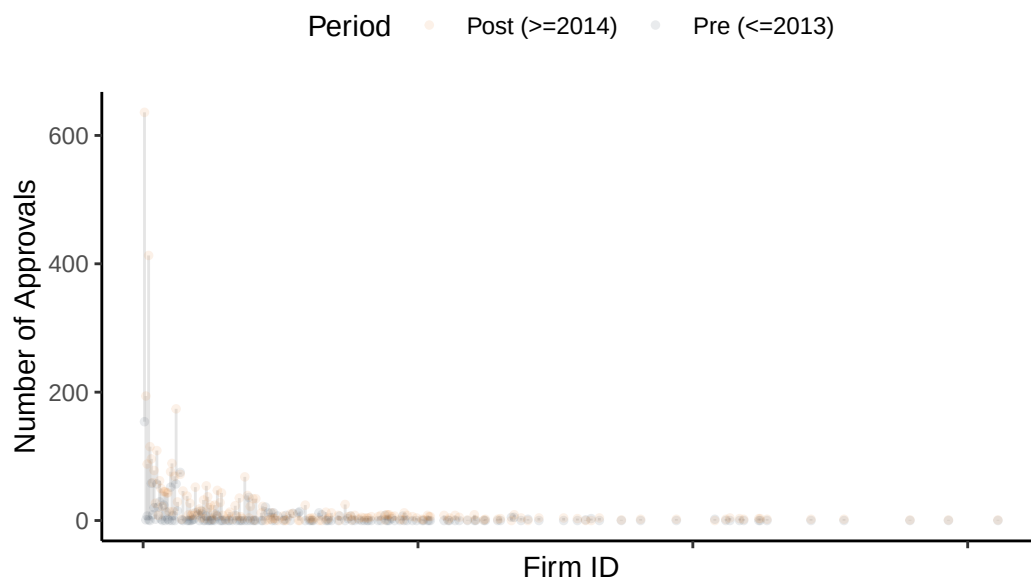
FDA Within-Firm Variation: ATC Level S



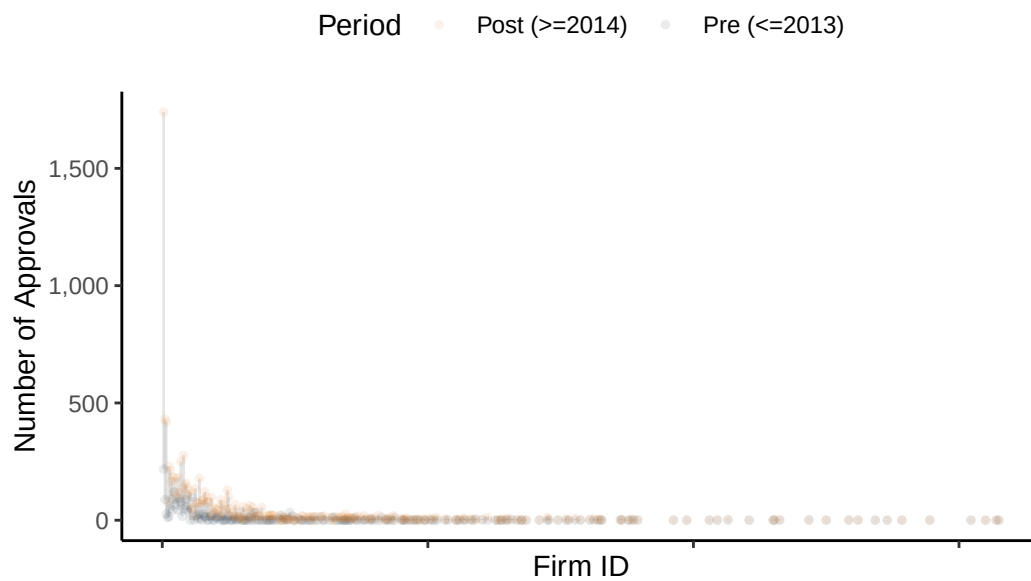
FDA Within-Firm Variation: ATC Level A



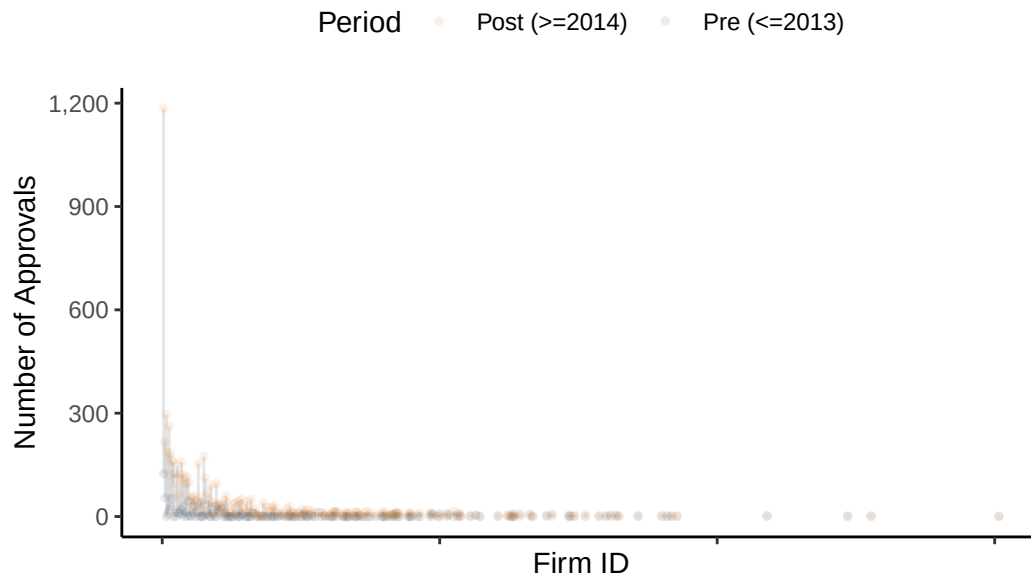
FDA Within-Firm Variation: ATC Level B



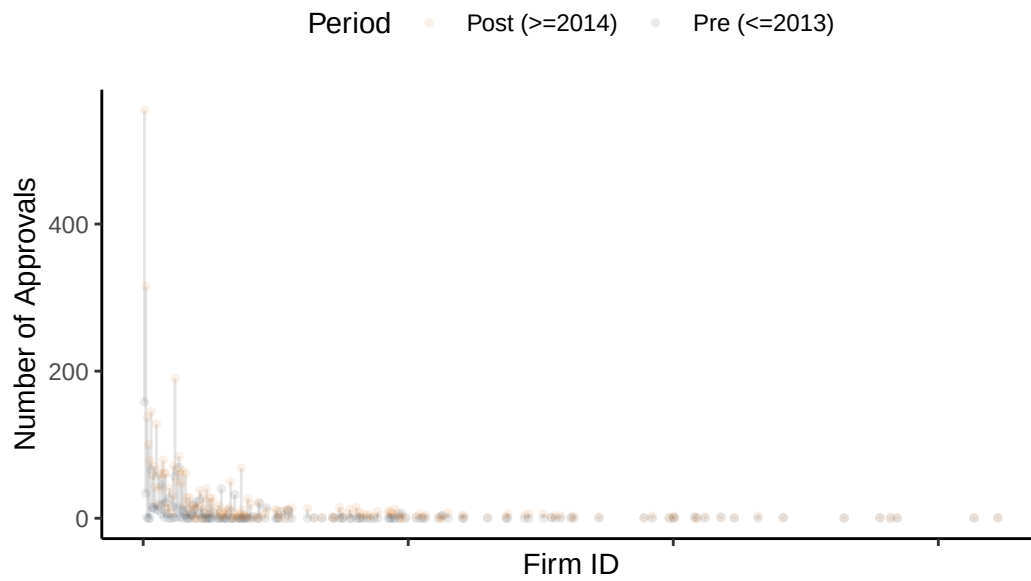
FDA Within-Firm Variation: ATC Level C



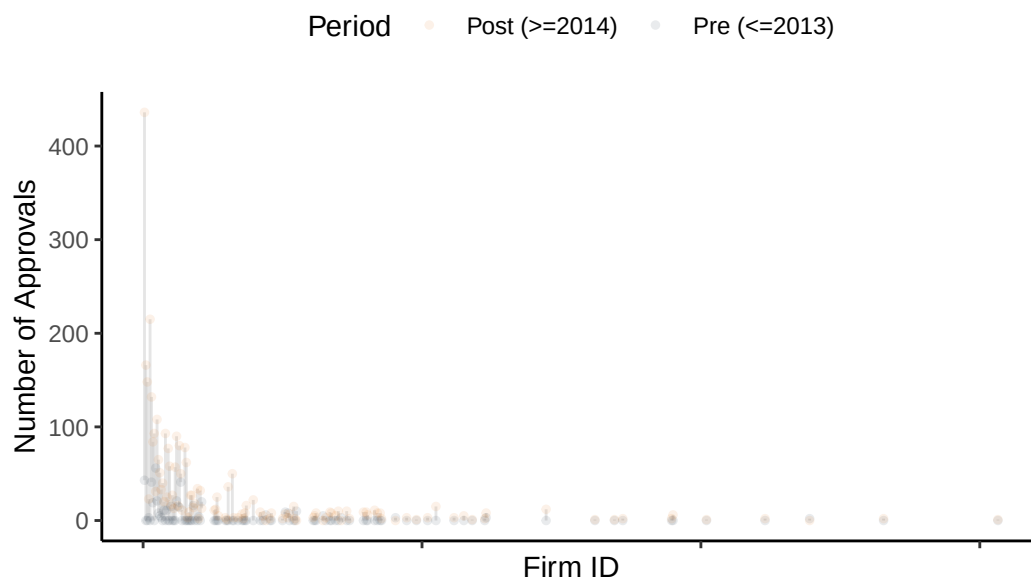
FDA Within-Firm Variation: ATC Level D



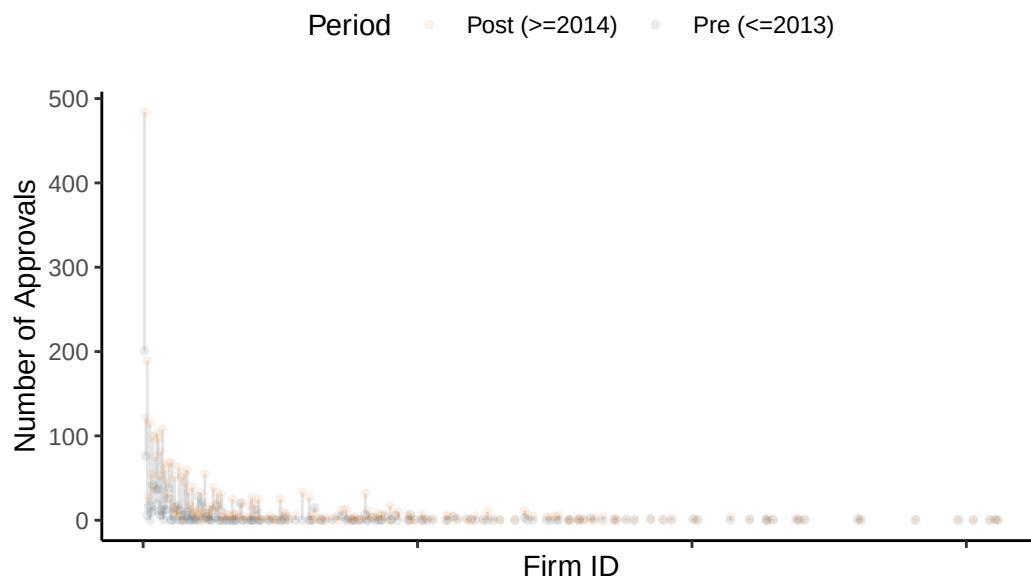
FDA Within-Firm Variation: ATC Level G



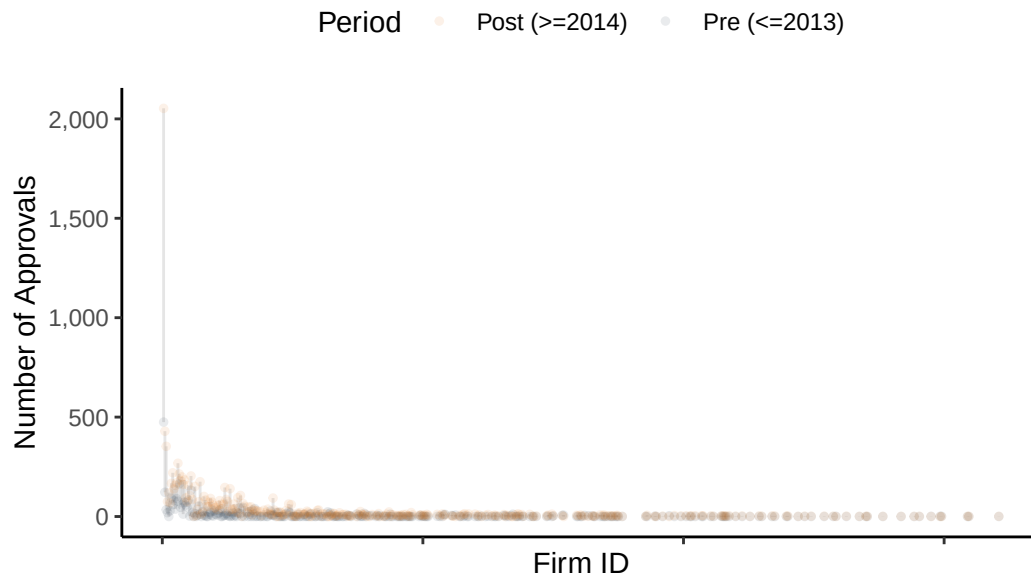
FDA Within-Firm Variation: ATC Level H



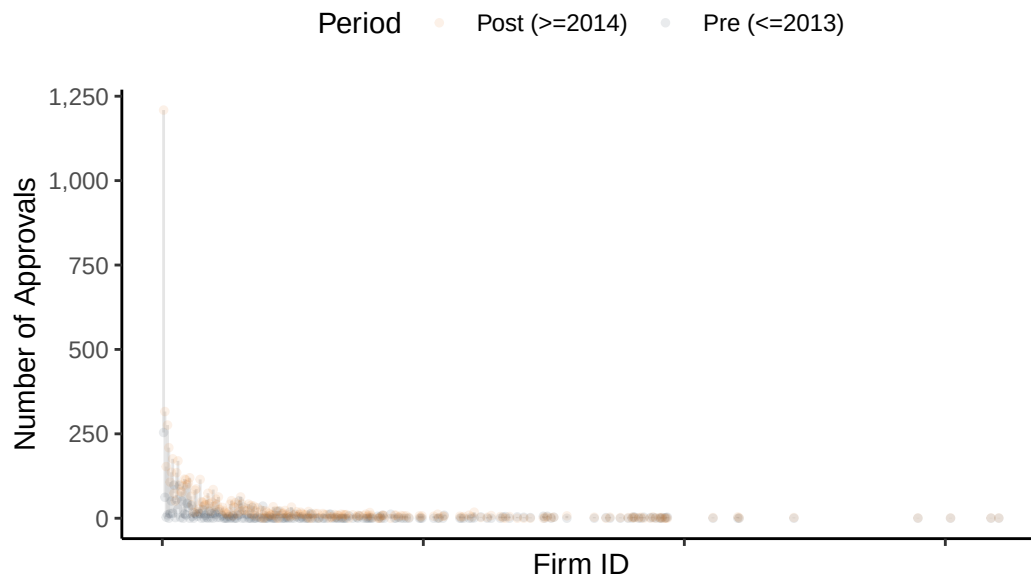
FDA Within-Firm Variation: ATC Level J



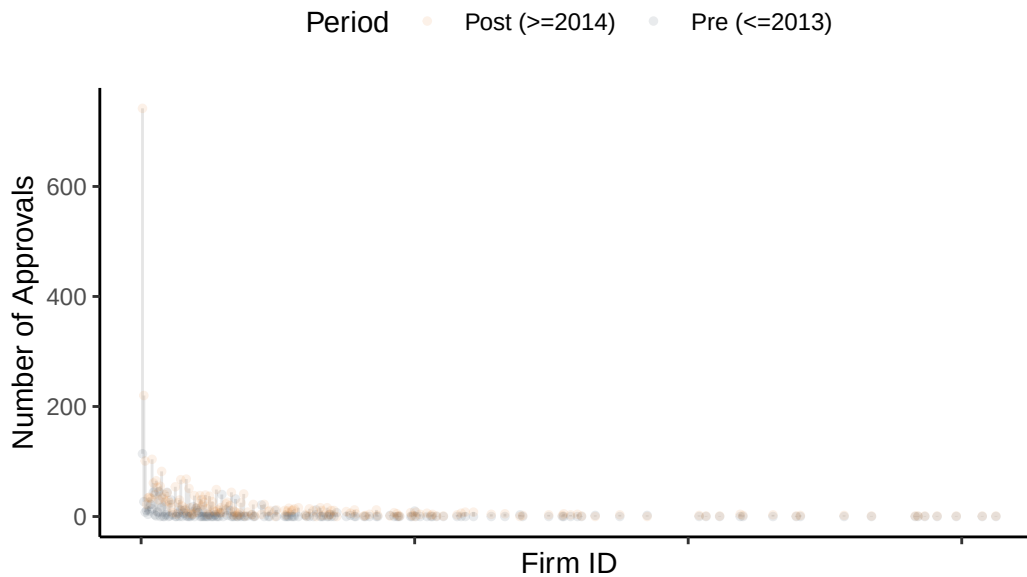
FDA Within-Firm Variation: ATC Level N



FDA Within-Firm Variation: ATC Level R



FDA Within-Firm Variation: ATC Level M



Patents

```
#| results: asis
#| fig-width: 10
#| fig-height: 6

patent_firm_atc_counts <- patents |>
  group_by(firm, atc_level1, post) |>
  summarise(patents = sum(patents, na.rm = TRUE), .groups = "drop") |>
  filter(patents > 0)

firm_index <- patent_firm_atc_counts |>
  group_by(firm) |>
  summarise(total_activity = sum(patents), .groups = "drop") |>
  arrange(desc(total_activity)) |>
  mutate(firm_id = row_number())

patent_plot_data <- patent_firm_atc_counts |>
  left_join(firm_index, by = "firm") |>
  pivot_wider(names_from = post, values_from = patents, values_fill = 0) |>
  rename(pre = `FALSE`, post_val = `TRUE`)
```

```

atc_list <- unique(patent_plot_data$atc_level1)

for (i in atc_list) {

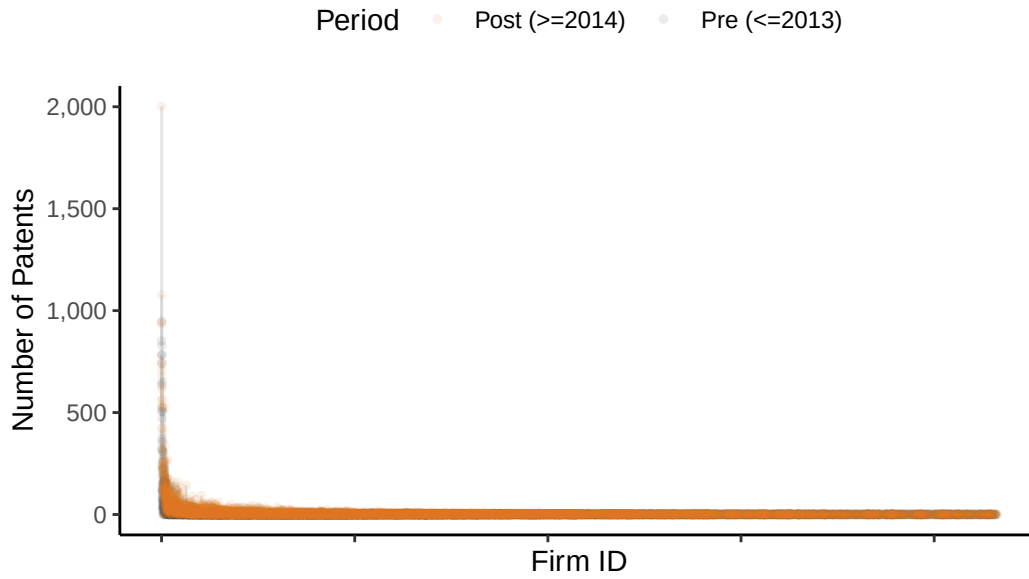
  single_atc_data <- patent_plot_data |>
    filter(atc_level1 == i)

  p <- ggplot(single_atc_data, aes(x = firm_id)) +
    geom_segment(aes(xend = firm_id, y = pre, yend = post_val),
      color = "grey80", alpha = 0.5) +
    geom_point(aes(y = pre, color = "Pre (<=2013)"), size = 1, alpha = 0.1) +
    geom_point(aes(y = post_val, color = "Post (>=2014)"), size = 1, alpha = 0.1) +
    scale_color_manual(values = c("Pre (<=2013)" = "#2c3e50", "Post (>=2014)" = "#e67e22")) +
    scale_y_continuous(labels = scales::comma) +
    labs(
      title = paste("Within-Firm Variation: ATC Level", i),
      x = "Firm ID",
      y = "Number of Patents",
      color = "Period"
    ) +
    theme_classic() +
    theme(
      axis.text.x = element_blank(),
      legend.position = "top"
    )

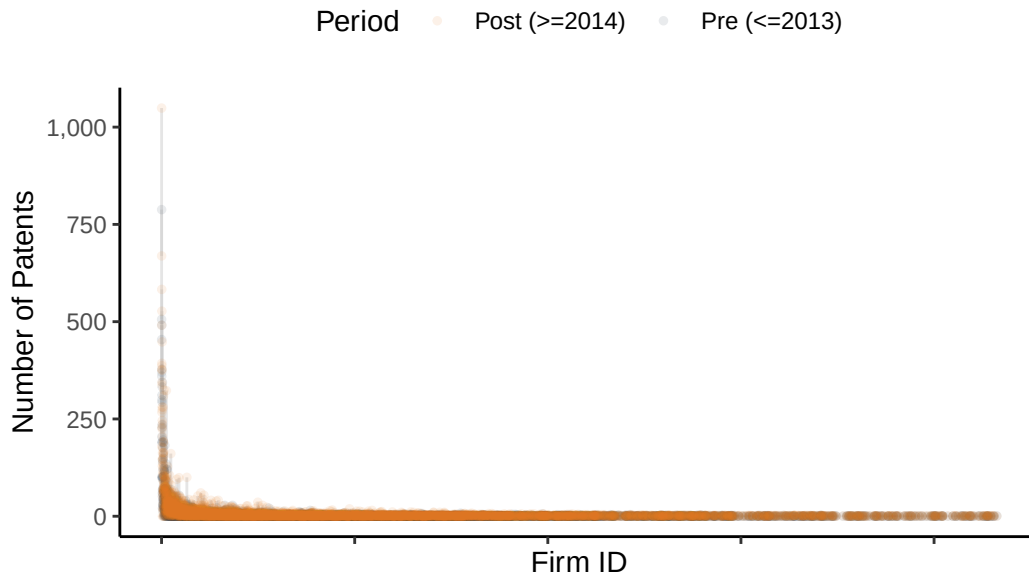
  print(p)
}

```

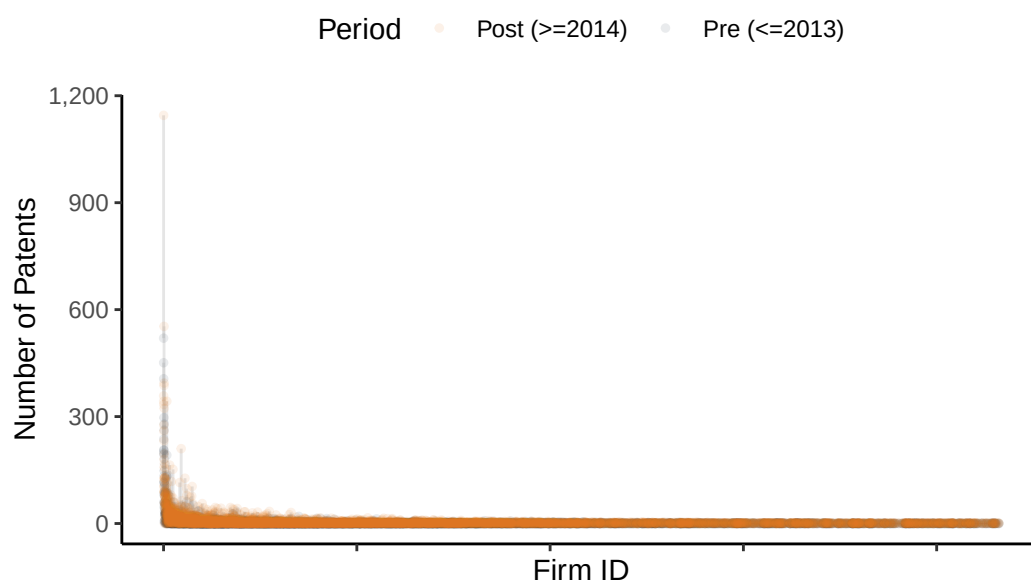
Within-Firm Variation: ATC Level A



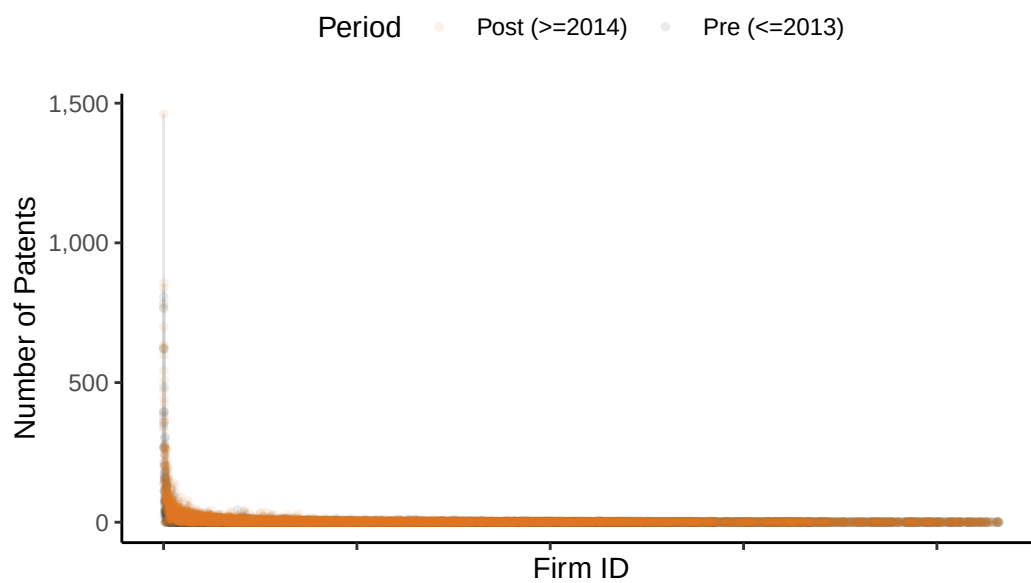
Within-Firm Variation: ATC Level C



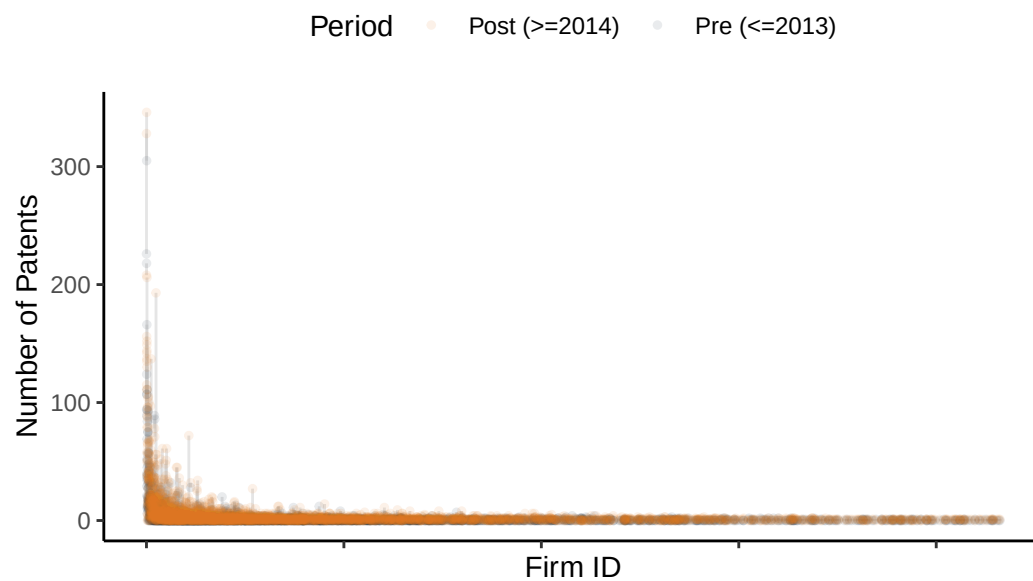
Within-Firm Variation: ATC Level D



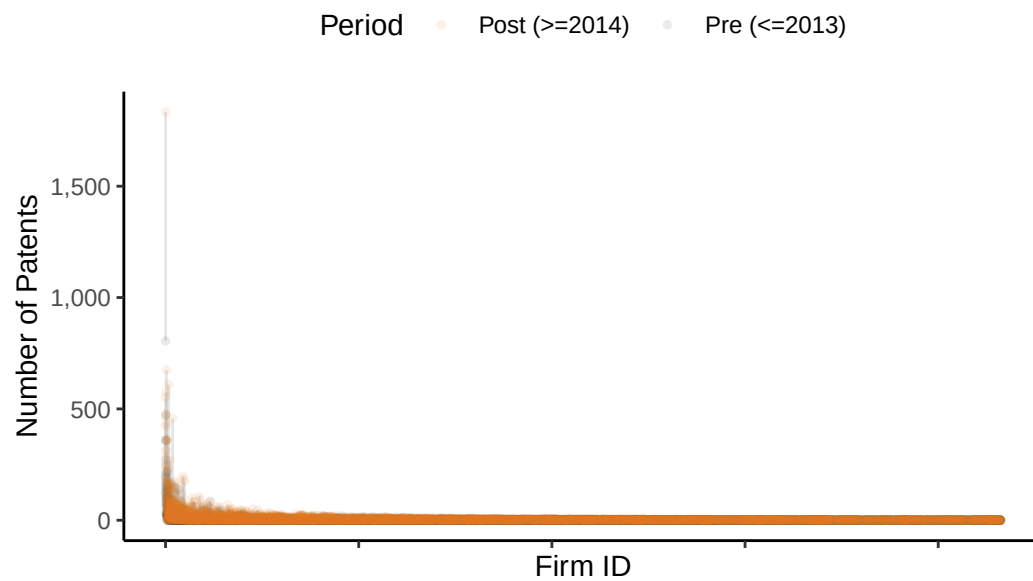
Within-Firm Variation: ATC Level M



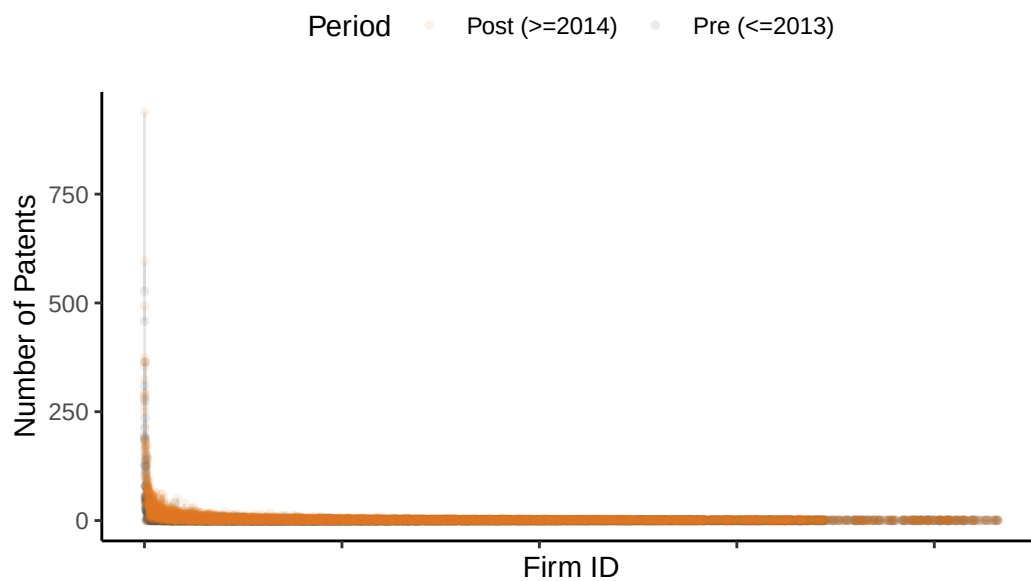
Within-Firm Variation: ATC Level B



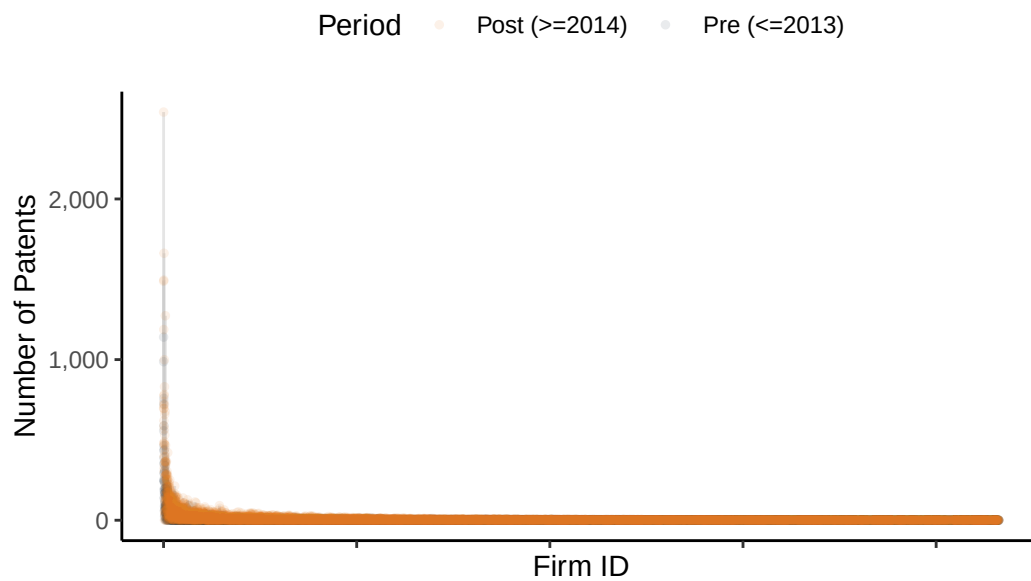
Within-Firm Variation: ATC Level J



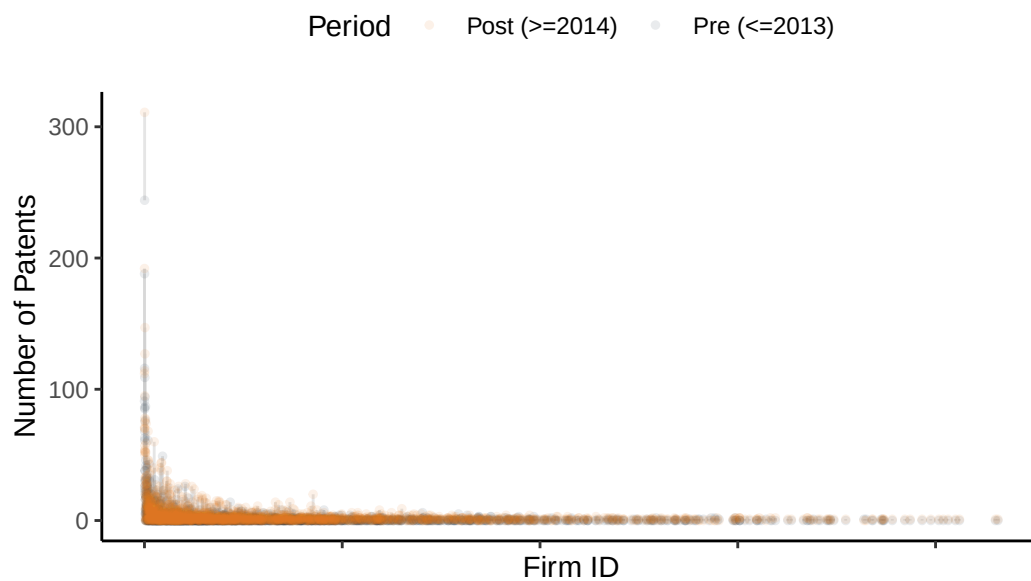
Within-Firm Variation: ATC Level V



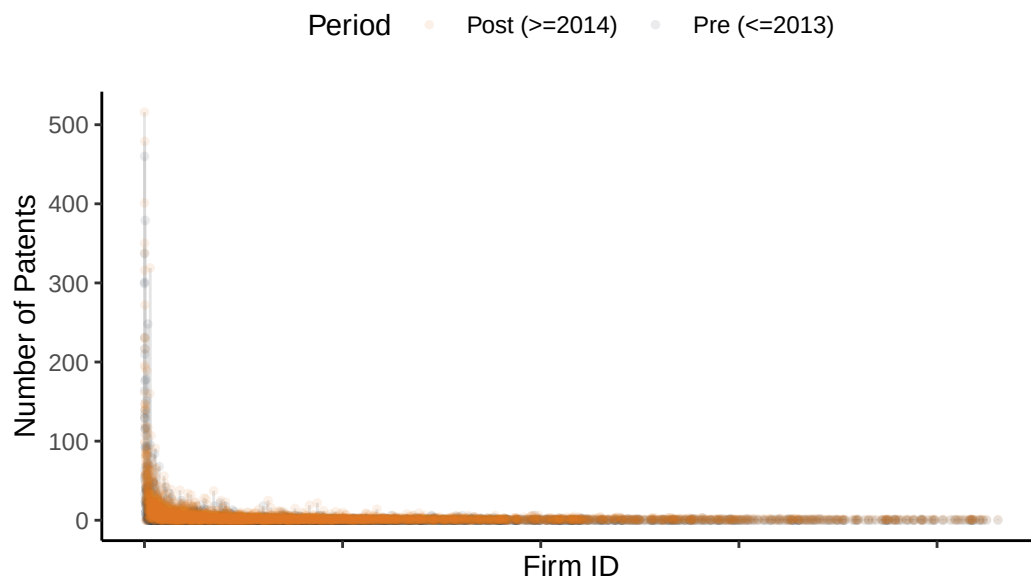
Within-Firm Variation: ATC Level L



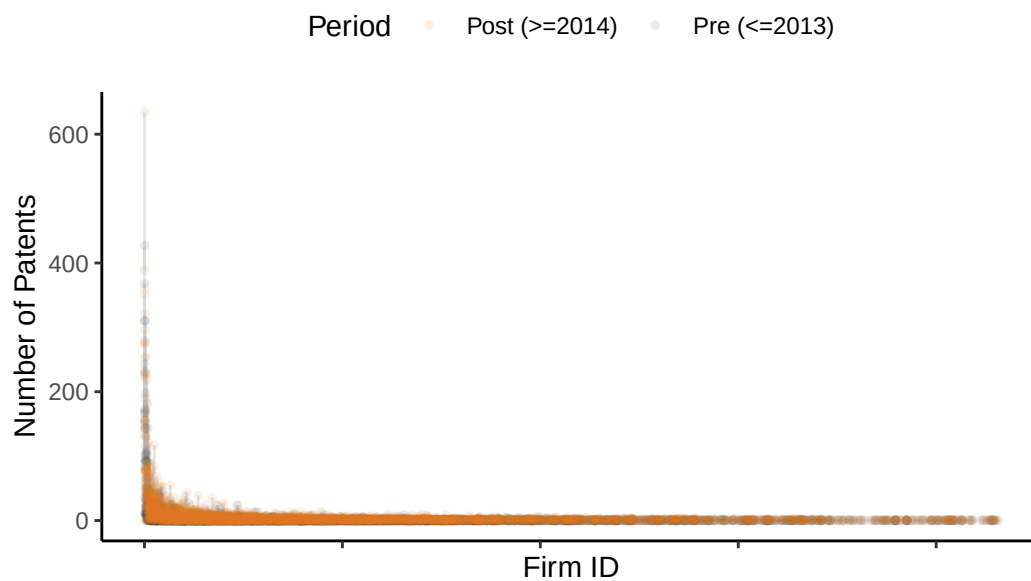
Within-Firm Variation: ATC Level H



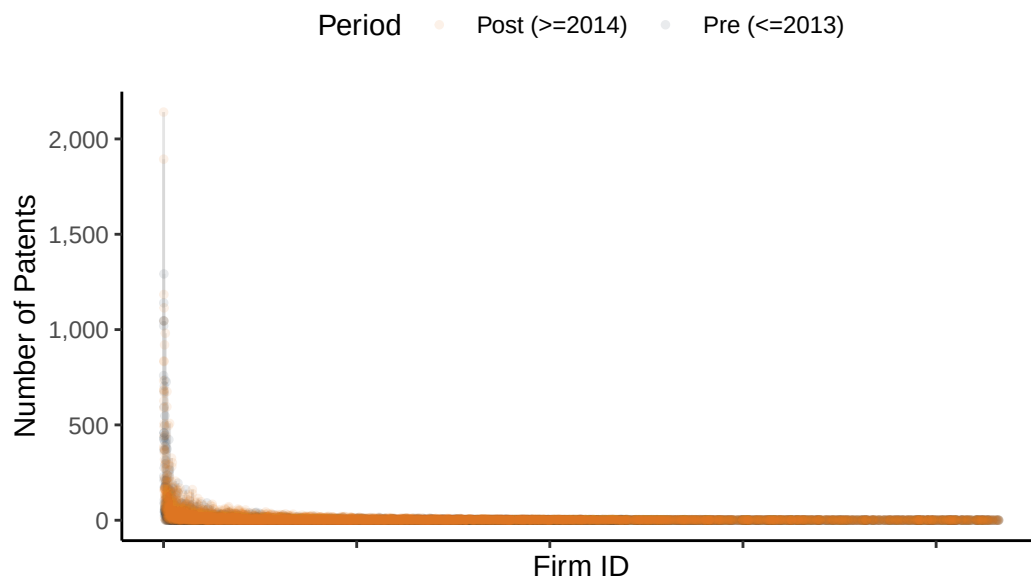
Within-Firm Variation: ATC Level R



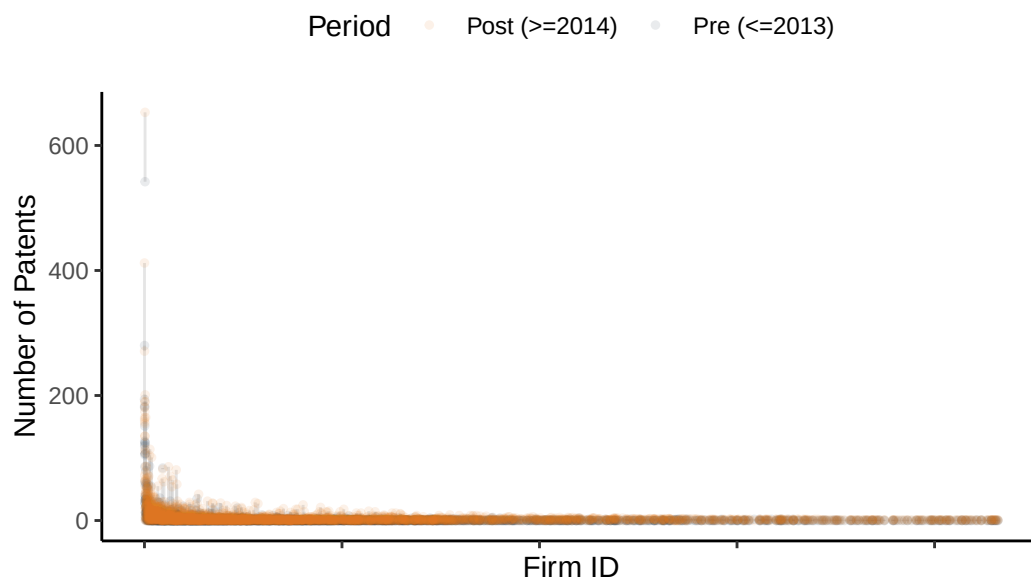
Within-Firm Variation: ATC Level G



Within-Firm Variation: ATC Level N



Within-Firm Variation: ATC Level S



Within-Firm Variation: ATC Level P

